

California Housing Price Prediction Using Machine Learning

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Project: Machine Learning Regression Model Comparison

1. Objective

The objective of this project was to compare the performance of three machine learning regression models—Linear Regression, Ridge Regression, and Decision Tree Regression—using the California Housing dataset. The models were evaluated using Root Mean Squared Error (RMSE) and R^2 Score to determine which model produced the most accurate predictions.

2. Methodology

Step 1: Import Libraries

The required Python libraries such as Pandas, NumPy, Matplotlib, Joblib, and Scikit-learn were imported. These libraries were used for data handling, visualization, preprocessing, model building, evaluation, and model saving.

Step 2: Load the Dataset

The California Housing dataset was loaded using the `fetch_california_housing()` function from Scikit-learn. The dataset was converted into a Pandas DataFrame for easier analysis.

Step 3: Data Exploration

The dataset was explored using the following functions:

- `df.head()` to view the first five rows.
- `df.shape` to check the number of rows and columns.
- `df.columns` to view all column names.
- `df.info()` to examine data types and missing values.
- `df.describe()` to obtain statistical information such as mean, standard deviation, minimum, and maximum values.

The exploration showed that all columns contained complete numerical data with no missing values.

Step 4: Feature Selection

The dataset was divided into:

- Features (X) – all input variables.
- Target (y) – MedHouseVal, which represents the house price.

Step 5: Train-Test Split

The dataset was divided into training and testing sets using the `train_test_split()` function.

- Training Data: 80%
- Testing Data: 20%

A `random_state` of 42 was used to ensure reproducible results.

Step 6: Feature Scaling

Feature scaling was performed using `StandardScaler`. Scaling was applied only to the Linear Regression and Ridge Regression models because these algorithms perform better when numerical features are on a similar scale. Decision Tree Regression was trained on the original dataset because tree-based algorithms do not require feature scaling.

Step 7: Model Training

Three regression models were trained:

- Linear Regression
- Ridge Regression
- Decision Tree Regression

Each model was trained using the training dataset and tested on the testing dataset.

Step 8: Model Evaluation

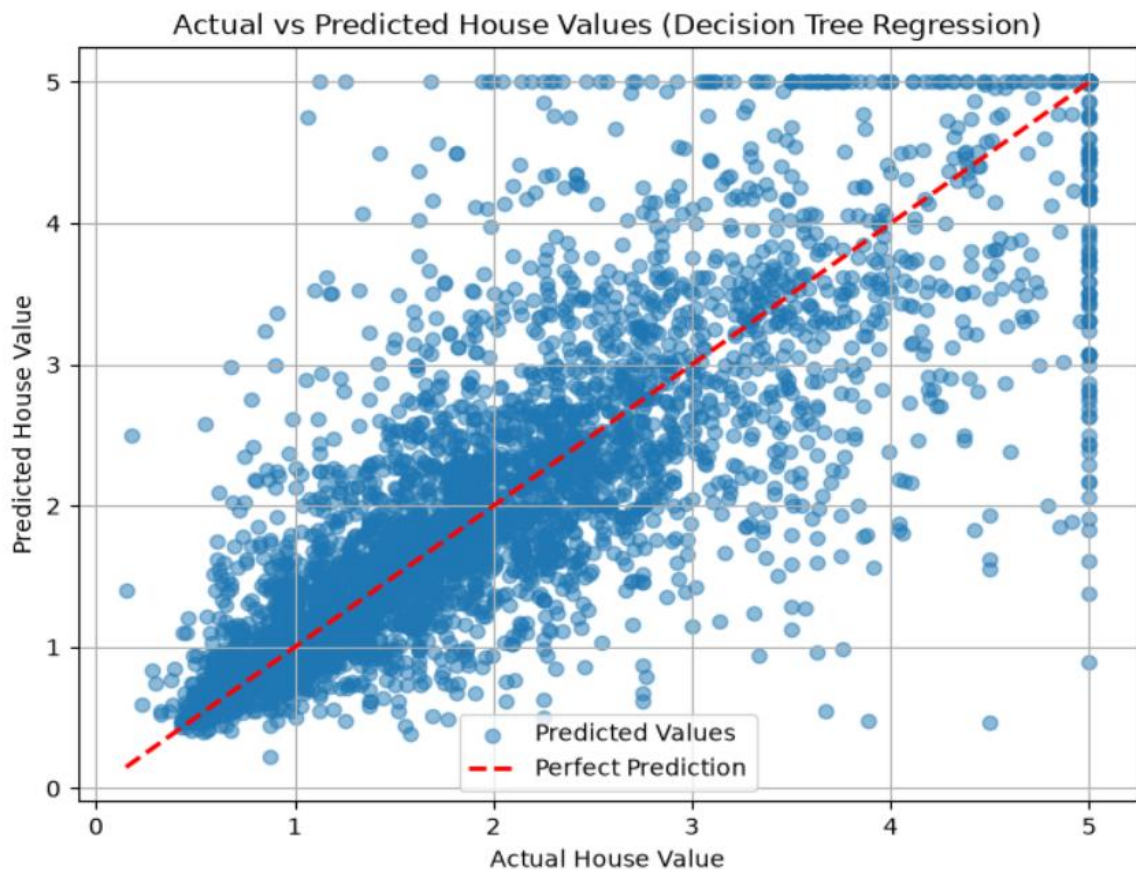
The models were evaluated using:

Root Mean Squared Error (RMSE): RMSE measures the average prediction error. Lower RMSE values indicate better model performance.

R² Score: R² Score measures how well the model explains the variation in the target variable. Higher R² values indicate better performance.

3.Results

MODEL	RMSE	R2 SCORE
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decesion Tree Regression	0.7030	0.6228



4.Conclusion:

The three regression models were successfully trained and evaluated using the California Housing dataset. Based on the evaluation metrics, Decision Tree Regression achieved the best performance with the lowest RMSE (0.7030) and the highest R^2 Score (0.6228). In comparison, Linear Regression and Ridge Regression produced almost identical results, indicating that regularization had little impact on this dataset.

To preserve the trained model for future use without retraining, the best-performing Decision Tree Regression model was saved using the Joblib library.

`joblib.dump(tree_model, "best_model.pkl")`

The saved model (best_model.pkl) can later be loaded using Joblib for making predictions on new data